ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** axis	Axis.
2	**(d)** hot	Warm.
3	**(b)** Sun and Earth	Blocks Sun.
4	**(d)** Sun	Sun.
5	**(c)** cold	Cool.
6	**(a)** Sun and Moon	Shadow on Moon.
7	**(b)** Earth	Earth.
8	**(d)** longer	Longer.
9	**(c)** new moon	Aligned.
10	**(d)** rotation	Rotation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Spinning on an axis (1); Causes day and night (1).	1+1
12	366 days (1); Every 4 years (1).	1+1
13	Moon blocks the Sun (1); At new moon (1).	1+1
14	Going round the Sun (1); Takes a year (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Day and night (1); Sunrise and sunset (1); Apparent motion of stars (1).	1+1+1
16	A star (1); Source of light (1); Hot (1).	1+1+1
17	24 hours (1); One rotation (1); Day and night (1).	1+1+1
	OR: Year (1); Seasons (1); Changing constellations (1).	1+1+1
18	Pinhole (1); Special glasses (1); Screen projection (1).	1+1+1
	OR: 365 days (1); One revolution (1); Plus quarter day (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earth is a sphere (1); Spins on its axis (1); Faces the Sun: day (1); Away: night (1); 24 hours cycle (1).	1+1+1+1+1
	OR: Full moon (1); Earth between (1); Shadow on Moon (1); Moon looks dark red (1); Safe to watch (1).	1+1+1+1+1
20	Axis tilted (1); Earth goes round the Sun (1); Summer: tilted towards (1); Winter: tilted away (1); Change in weather (1).	1+1+1+1+1
	OR: Do not look at the Sun (1); Use pinhole (1); Use safe filters (1); Protect children (1); Follow experts (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** shorter	Shorter.
2	**(c)** full moon	Aligned.
3	**(a)** tilt of Earth	Tilt.
4	**(b)** 0° latitude	Middle.
5	**(d)** solar eclipse	Harmful.
6	**(b)** east	East.
7	**(a)** tilted	23.5°.
8	**(b)** naked eyes	Safe.
9	**(d)** west	West.
10	**(c)** a satellite	Natural.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A natural satellite (1); Of the Earth (1).	1+1
12	Earth's shadow on Moon (1); At full moon (1).	1+1
13	Rotation (1); Of the Earth (1).	1+1
14	No Moon seen (1); Dark (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New moon (1); Half moon (1); Full moon (1).	1+1+1
16	Summer (1); Winter (1); Rainy (1).	1+1+1
17	Quarter days (1); Add up to one day (1); Calendar fits (1).	1+1+1
	OR: 21 March (1); 23 September (1); Equal day and night (1).	1+1+1
18	Rotation: spin (1); Revolution: orbit (1); 24 hours vs 1 year (1).	1+1+1
	OR: We see different parts (1); It orbits (1); Reflects light (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	New moon (1); Crescent (1); Half (1); Gibbous (1); Full moon (1).	1+1+1+1+1
	OR: Year is 365.25 days (1); Quarter day extra (1); Four years make a day (1); Added to February (1); 366 days (1).	1+1+1+1+1
20	Rotates (1); Revolves (1); Tilted (1); Has one Moon (1); Has life (1).	1+1+1+1+1
	OR: Summer is hot (1); Winter is cold (1); Long days in summer (1); Short days in winter (1); Different clothes (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that seasons are caused by the Earth being closer to or farther from the Sun; they are caused by the **tilt of the axis** and revolution.
- Mixing up rotation (day and night, 24 hours) with revolution (year, $365\frac{1}{4}$ days).
- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
- Writing that the Moon gives its own light; it only **reflects** sunlight.
- Thinking the phases of the Moon are caused by the Earth's shadow; they are due to how much of the sunlit half we see.
- Writing that it is summer in both hemispheres at the same time; the seasons are **opposite** in the two hemispheres.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** about a month	29.5 days.
2	**(b)** 24 hours	24 h.
3	**(a)** reflected sunlight	Reflects.
4	**(a)** always	Same side.
5	**(c)** 365 days	365.
6	**(a)** dark	Moon not visible.
7	**(c)** the Moon has no light of its own	Reflects.
8	**(d)** 366 days	One more.
9	**(d)** bright	Whole disc.
10	**(a)** one revolution	Around Sun.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Northern (1); Southern (1).	1+1
12	Tilt of the axis (1); And revolution (1).	1+1
13	Whole disc seen (1); Bright (1).	1+1
14	Equator (1); Divides the Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	21 June (1); 22 December (1); Longest and shortest (1).	1+1+1
16	Sun's rays are direct (1); Longer days (1); More heat (1).	1+1+1
17	Use a globe (1); Use a torch (1); Rotate the globe (1).	1+1+1
	OR: Satellite (1); No air (1); Craters (1).	1+1+1
18	Slanting rays (1); Shorter days (1); Less heat (1).	1+1+1
	OR: Light at night (1); Calendar (1); Festivals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Satellite (1); No air (1); No water (1); Craters (1); Weak gravity (1).	1+1+1+1+1
	OR: Take a globe (1); Take a torch (1); Rotate the globe (1); Lit part day (1); Dark part night (1).	1+1+1+1+1
20	New moon (1); Moon between Sun and Earth (1); Shadow on Earth (1); Dark for some time (1); Unsafe to watch (1).	1+1+1+1+1
	OR: Light (1); Heat (1); Calendar (1); Festivals (1); Seasons (1).	1+1+1+1+1

Mistakes to avoid:

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- Interchanging eclipses: **solar** eclipse = Moon between Sun and Earth (new moon); **lunar** eclipse = Earth between Sun and Moon (full moon).
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